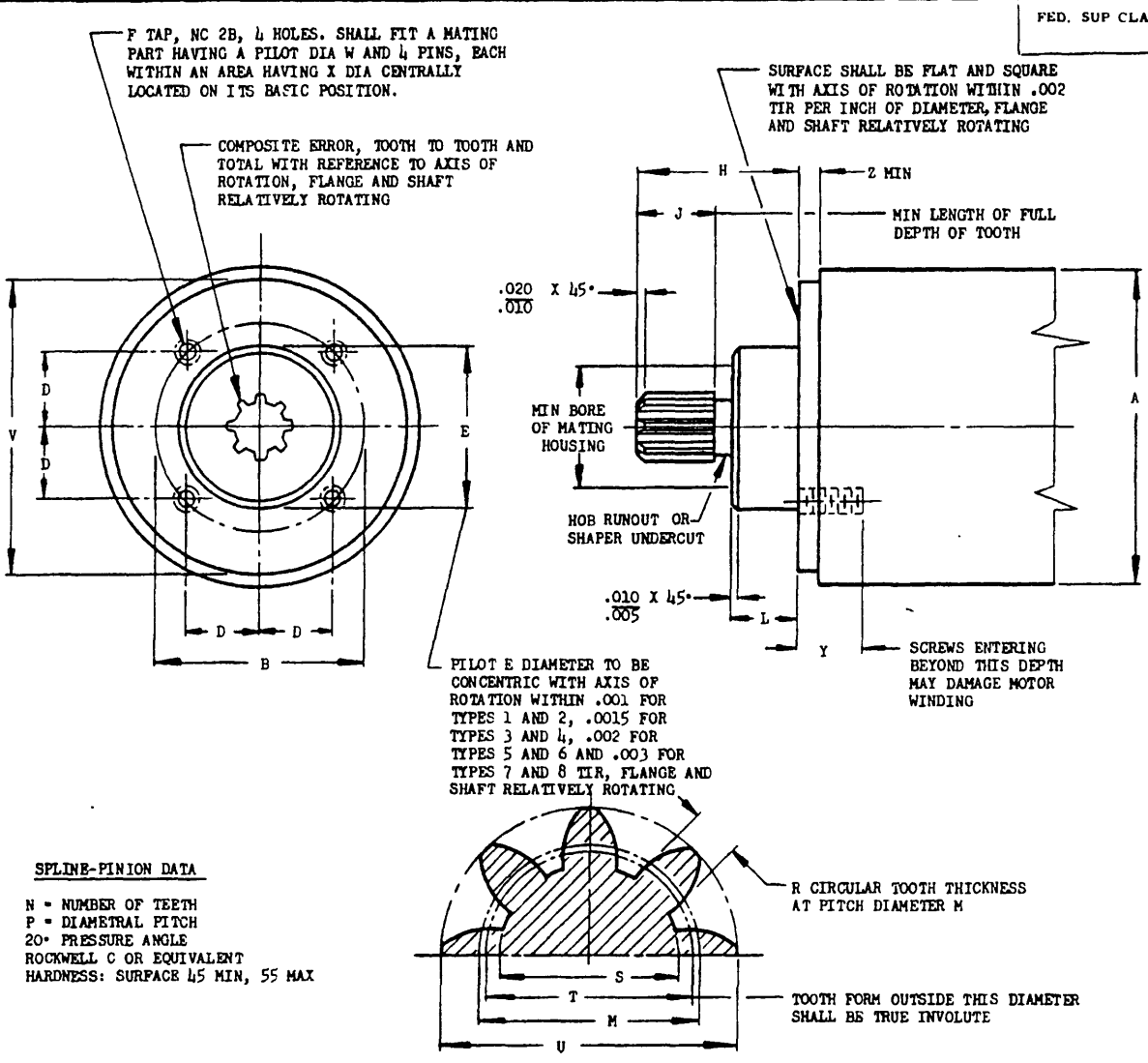
FED. SUP CLASS



SPLINE-PINION DATA

N = NUMBER OF TEETH
P = DIAMETRAL PITCH
20° PRESSURE ANGLE
ROCKWELL C OR EQUIVALENT
HARDNESS: SURFACE 45 MIN, 55 MAX

SPLINE-PINION DETAIL

(a) A IS GREATEST DIAMETER OR DIMENSION OF ACCESORY EXCLUSIVE OF ATTACHED DEVICES.

(b) OUTSIDE DIAMETERS OF SMALL PINIONS ARE ENLARGED TO AVOID UNDERCUT. TO MAINTAIN STANDARD CENTER DISTANCE, THE GEAR DIAMETERS MUST BE DECREASED BY THE AMOUNT OF THE PINION ENLARGEMENT GIVEN AT Q. IF MATING GEARS ARE MADE WITH STANDARD TOOTH PROPORTIONS, THE CENTER DISTANCES MUST BE INCREASED APPROXIMATELY 1/2 Q.

(c) R MAX HAS BEEN REDUCED FROM THEORETICAL BY TOTAL COMPOSITE ERROR OF PINION ONLY. RANGE RESULTS FROM SIZE VARIABLE ONLY.

(d) OTHER MASTER GEARS MAY BE USED IF CENTER DISTANCE IS PROPERLY RECALCULATED.

(e) RANGE OF CENTER DISTANCE IS THAT RESULTING FROM RMS SUMMATION OF ALL VARIABLES IN THE STRUCTURE.

SIZE DETERMINATION OF MATING GEAR MUST RECOGNIZE FURTHER VARIATIONS IN ITS SUPPORTING STRUCTURE.

SPLINE-PINION DIMENSIONS ARE BASIC MAXIMUM.

PINION DIMENSIONS ARE ADEQUATE FOR TYPE SHOWN BUT ALL PINIONS SHALL BE CHECKED FOR APPLICATION.

REMOVE ALL BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

NOMINAL USE OF THIS DRIVE FOR AC OR DC ELECTRICAL MOTORS.

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APPROVED 1 May 58 REVISED

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33600 (ASG)
Navy - BuAer Air Force	A - N - AF -	DRIVE, MINIATURE, POWER MOTOR WITH INVOLUTE SPLINE - PINION	
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 1 OF 4

DD FORM 672-1

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DATA FOR BASIC DRIVE

TYPE	FTIN SIZE	U DIA MAX	U DIA MIN	V	W ±.0001 DIA	X ±.0001 DIA	Y MIN DEPTH	Z MIN	MEAS WIRE DIA	MEAS OVER 2 WIRES MAX	MEAS OVER 2 WIRES MIN	(d) MASTER GEAR NO. OF TEETH	(d) MASTER GEAR TOOTH THK	CENTER DISTANCE (e) WITH MASTER THEO	CENTER DISTANCE (e) WITH MASTER MAX	CENTER DISTANCE (e) WITH MASTER MIN	COMPOSITE ERROR TOOTH TO TOOTH	COMPOSITE ERROR TOTAL	MIN BORE OF MATING HOUSING
1	06	.0921	.0902	.516	.3130	.0444	.109	.047	.01600	.0955	.0952	120	.01309	.5276	.5375	.5367	.0003	.0006	.280
2	08	.1148	.1130	.630	.3755	.0532	.156	.062	.02000	.1193	.1190	96	.01636	.5470	.5468	.5460	.0004	.0008	.315
3	09	.1240	.1222	.750	.5005					.1277	.1274			.5517	.5515	.5505			.440
4	11	.1530	.1508	.937	.5630	.0730	.203		.02667	.1591	.1587	72	.02182	.5626	.5624	.5612	.0005	.0011	.524
5	13	.1722	.1700	1.125	.6255	.0784			.03000	.1791	.1787	64	.02454	.5704	.5703	.5689			.565
6	15	.1965	.1943	1.375	.7505			.078	.03429	.2046	.2042	56	.02805	.5804	.5802	.5787	.0006	.0015	.670
7	18	.2294	.2270	1.625	.8755	.0966	.250		.04000	.2388	.2383	48	.03272	.5938	.5937	.5918			.795
8	20	.2474	.2450	1.875	1.0211			.094		.2554	.2549			.6031	.6028	.6009	.0007	.0020	.920

TYPE	SIZE	(a) A BASIC	(a) A MAX	B DIA REF	D BASIC	F TAP	H MAX	H MIN	J MIN	L MAX	L MIN	M ±PD THEO	N	P	(b) Q THEO	(c) R MAX EFF	(c) R MIN ACT	S DIA THD	T MAX DIA
1	06	.625	.635	.432	.1530	0-80	.321	.306	.125	.073	.067	.0667	8	120	.0089	.0162	.0158	.0515	.0589
2	08	.750	.760	.522	.1845	1-64	.356	.341	.145	.082	.076	.0833	9	96	.0111	.0202	.0198	.0654	.0736
3	09	.875	.891	.647	.2289		.378	.358				.0938			.0099	.0199	.0194	.0746	.0828
4	11	1.062	1.078	.767	.2712	3-48	.458	.438	.180	.101	.095	.1111		72	.0118	.0270	.0264	.0886	.0982
5	13	1.250	1.270	.919	.3248	4-40	.503	.473	.197			.1250	8	64	.0166	.0305	.0298	.1001	.1104
6	15	1.500	1.520	1.115	.3942		.584	.554	.220			.1429		56	.0190	.0348	.0340	.1150	.1262
7	18	1.750	1.781	1.365	.4826	6-32	.656	.616	.250	.145	.135	.1667		48	.0222	.0407	.0398	.1348	.1472
8	20	2.000	2.031	1.615	.5710	1.0236	1.0216	.673	.633	.163	.153	.1875	9		.0197	.0397	.0387	.1532	.1656

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APPROVED 1 May 58 REVISED

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CUSTODIANS	OTHER INT.
Navy - BuAer Air Force	A - N - AF -

MILITARY STANDARD

DRIVE, MINIATURE, POWER MOTOR WITH INVOLUTE SPLINE - PINION

MS33600 (ASG)

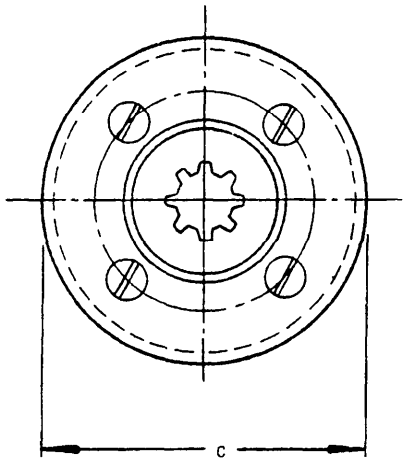
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF 4
NONE		

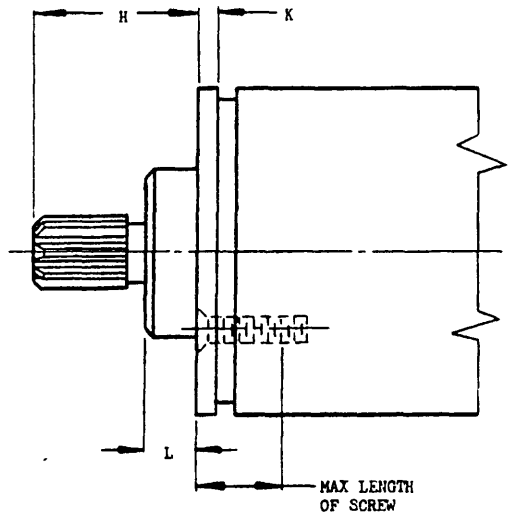
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FED SUP CLASS.





DATA FOR BASIC DRIVE WITH ROUND PLATE ATTACHMENT

TYPE	FIIN	SIZE	C +.003 -.010 DIA	H		K		L REF MEAN	MAXIMUM LENGTH OF SCREW
				MAX	MIN	MAX	MIN		
11		06	.625	.291	.276	.033	.027	.040	.1250
12		08	.750	.320	.305	.039	.033	.042	.1875
13		09	.875	.336	.316	.045	.039	.056	
14		11	1.062	.407	.387	.056	.046	.047	.2500
15		13	1.250	.443	.413	.065	.055	.038	
16		15	1.500	.512	.482	.077	.067	.069	.3125
17		18	1.750	.571	.531	.090	.080	.056	
18		20	2.000	.576	.536	.102	.092	.060	

SEE SHEET 2 FOR DIMENSIONS NOT SHOWN.

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APPROVED 1 May 58 REVISED

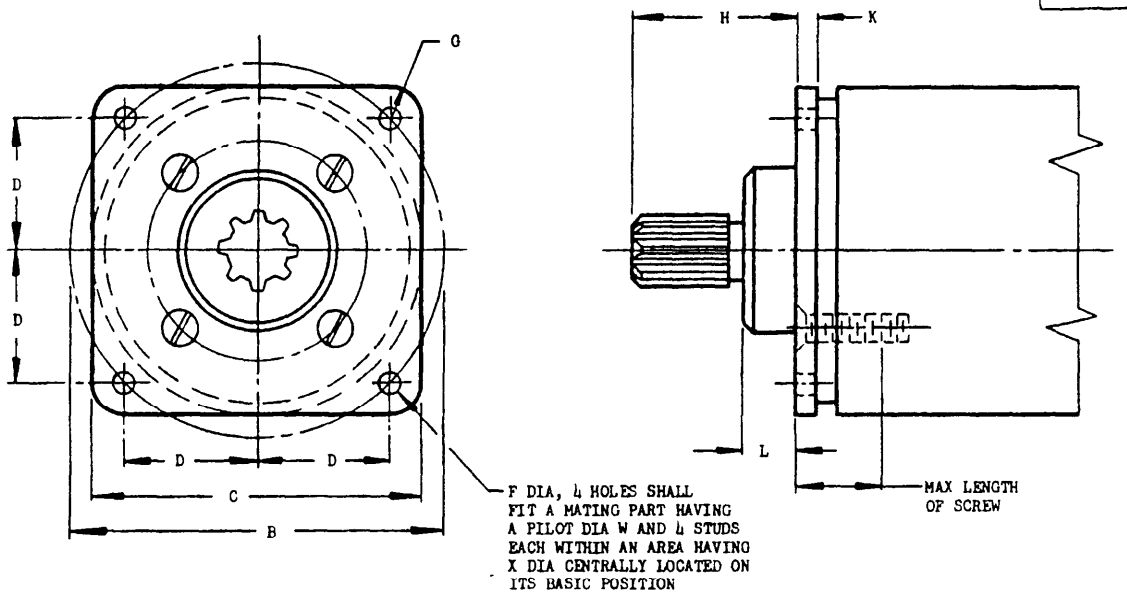
CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33600 (ASG)
Navy - BuAer Air Force	A - h - AF -	DRIVE, MINIATURE, POWER MOTOR WITH INVOLUTE SPLINE - PINION	
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 3 OF 4

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FED SUP CLASS



DATA FOR BASIC DRIVE WITH SQUARE PLATE ATTACHMENT

TYPE	FIN	SIZE	B DIA REF	C MAX	C MIN	D BASIC	F DIA MAX	F DIA MIN	G +.000 -.020 RAD	H MAX	H MIN	K MAX	K MIN	L REF MEAN	W ±.0001 DIA	X DIA	MAXIMUM LENGTH OF SCREW
21		06	.736	.635	.625	.260	.072	.069	.050	.291	.276	.033	.027	.040	.3130	.0670	.1250
22		08	.883	.760	.750	.312	.089	.085	.062	.320	.305	.039	.033	.042	.3755	.0821	.1875
23		09	1.016	.891	.875	.359			.078	.336	.316	.045	.039	.056	.5005		
24		11	1.225	1.078	1.062	.433	.103	.099	.094	.407	.387	.056	.046	.047	.5630	.0960	.2500
25		13	1.463	1.270	1.250	.517	.140	.134	.109	.443	.413	.065	.055	.038	.6255	.1375	
26		15	1.756	1.520	1.500	.621	.170	.163	.125	.512	.482	.077	.067	.069	.7505	.1554	.3125
27		18	2.021	1.781	1.750	.715			.156	.571	.531	.090	.080	.056	.8755		
28		20	2.271	2.031	2.000	.803			.203	.576	.536	.102	.092	.060	1.0211		

SEE SHEET 2 FOR DIMENSIONS NOT SHOWN.

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APPROVED 1 May 58 REVISED

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33600 (ASG)
Navy - BuAer Air Force	A - N - AF -	DRIVE, MINIATURE, POWER MOTOR WITH INVOLUTE SPLINE - PINION	
PROCUREMENT SPECIFICATION NONE		SUPERSEDES:	SHEET 4 OF 4

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